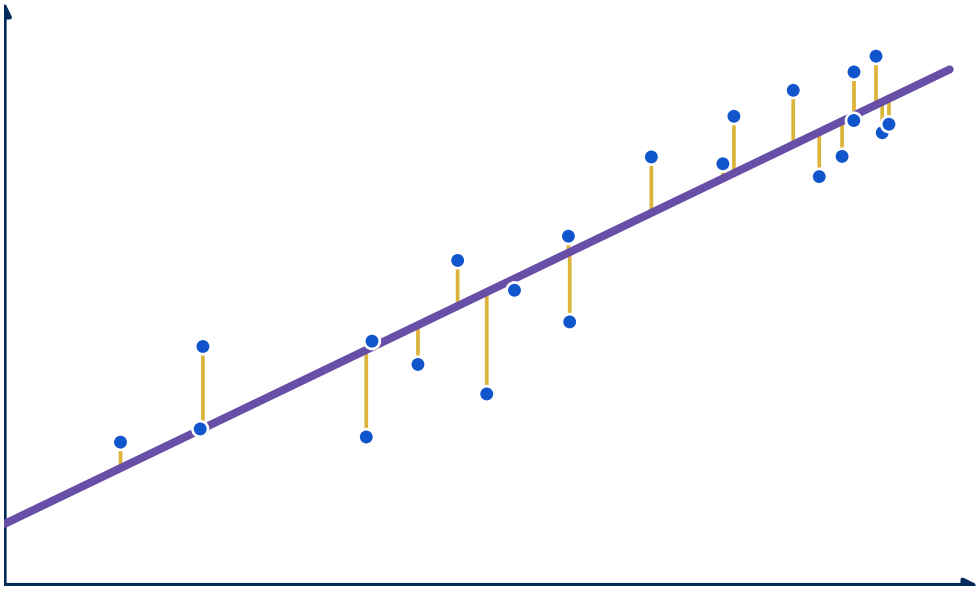


UNIT 02 · CHAPTER 2

# Linear Regression

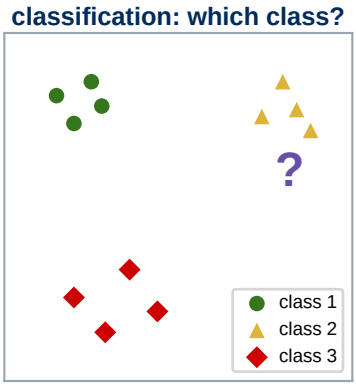
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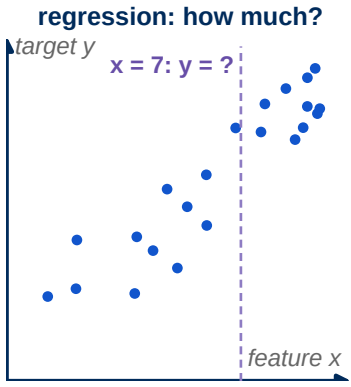
William Theisen · University of Notre Dame

# Classification and regression

Our **previous models**, **k-Nearest Neighbors** and **Decision Trees**, were built for **classification**.

A **classification task** is one in which we try to assign a _____ to a sample.




In **regression tasks**, we try and assign a _____ to each sample.

**Regression** is another form of _____ _____.

**Quick check (old exam question):** is a regression task supervised or unsupervised learning?

# Linear regression

A **linear regression** is a technique to fit a line to our data, allowing us to make predictions.

## Regression vocabulary

Unfortunately, because linear regression is so old, it **has a lot of vocab specific to it**:

[2.1] a line with one feature

$$\hat{y} = w_1 x_1 + w_0$$

- $\hat{y}$: _____
- $w_1$, $w_0$: _____
- $x_1$: _____

# Matrix notation

We can represent basically everything as **matrices** and **matrix operations**.

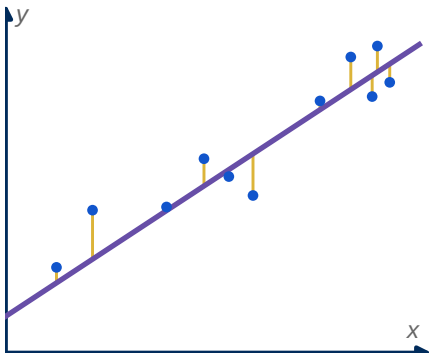
# Multiple features

It's rare that we have datasets with only a single feature. The **iris dataset**, a toy dataset, has 4!

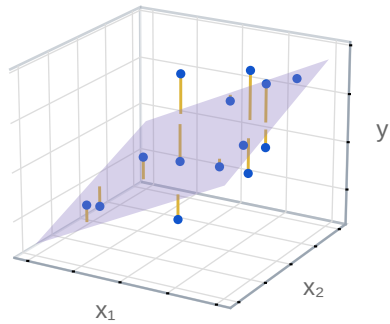
When we perform a **linear regression on** a dataset with **more than one feature**, we call this a _____.

## Hyperplanes

**1 feature: a line**




**2 features: a plane**

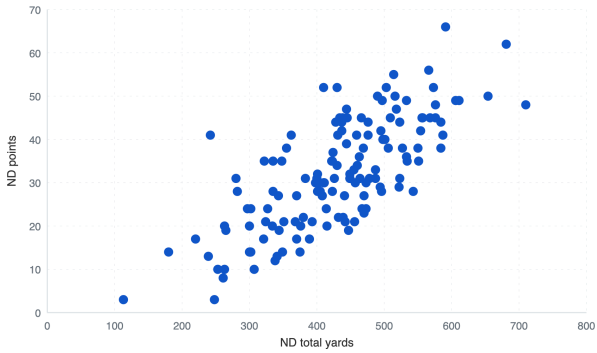



With one feature the model is a line, with two it's a _____ and with more it's a _____.

**Quick check (old exam question):** what would the equation look like for a multiple linear regression with three features?

# Fitting our line

How do we actually find the line that "fits" the best?




Draw the line you think fits best. Which games is it furthest from?

# Optimization problems

We need some way to **measure how well our model fits** our data. We can call this a ____ _____.

For **linear regression** we often use **SSE** or ____.

sum of squared errors

$$[2.3] \quad SSE(w) = \sum_i (y_i - \hat{y}_i)^2$$

mean squared error

$$[2.4] \quad MSE(w) = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

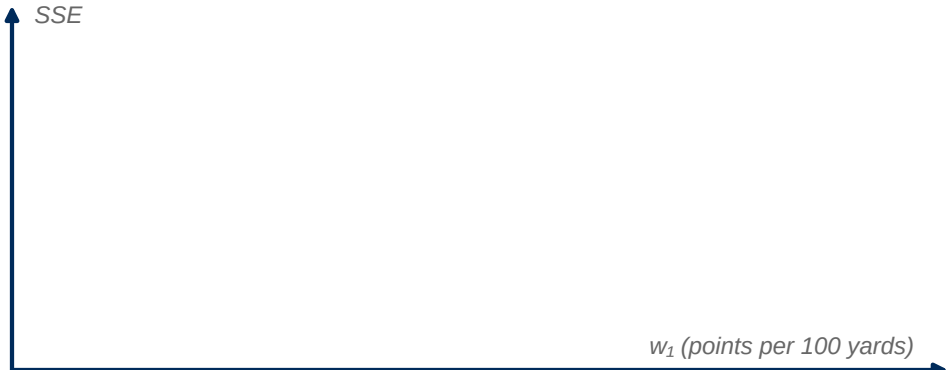
# Finding our weights

So given a **loss function**, our **goal** is to _____ it.

When would a **loss function** be **minimized**? When the _____ is ____.

## Visualizing loss

We can plot the loss function with respect to **w**:



# Ordinary least squares

We can **solve for the gradient equaling 0**:



This is called **Ordinary Least Squares (OLS)**.

# Worked example: OLS by hand

| game | yards | $x_1$ | y (points) |
|---|---|---|---|
| 2019 Michigan | 180 | 2 | 14 |
| 2012 Michigan State | 300 | 3 | 20 |
| 2013 Purdue | 400 | 4 | 31 |
| 2019 Boston College | 501 | 5 | 40 |
| 2017 Temple | 606 | 6 | 49 |

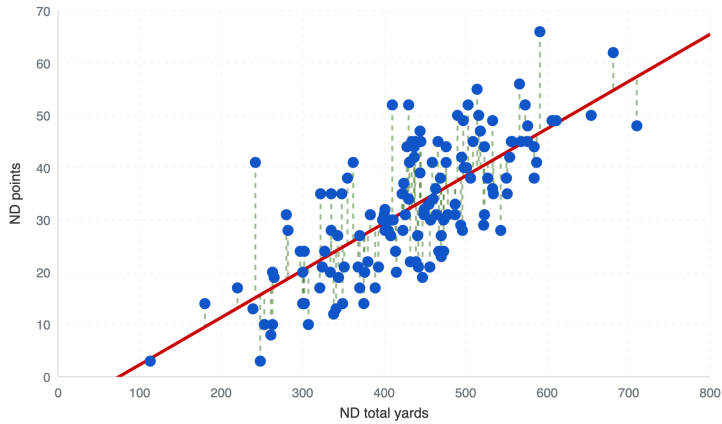
Give $X$ a **column of 1s** for the intercept, then follow the **OLS** recipe.

# Reading the weights

So $\hat{y} = 9.0\,x_1 - 5.2$: about _____ _____.

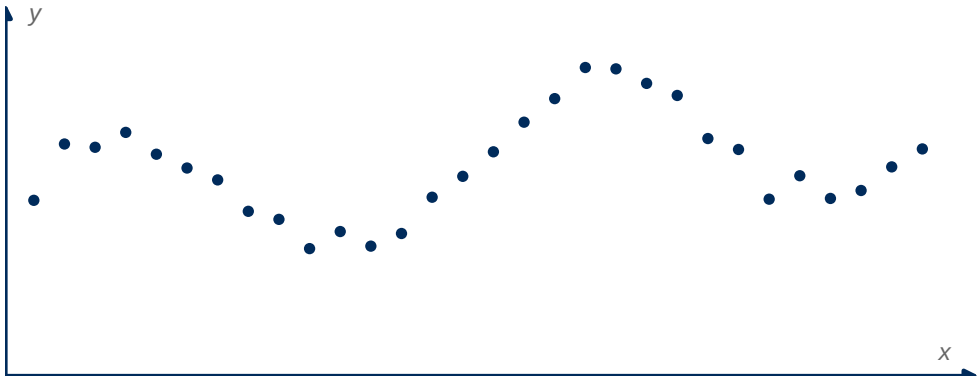
**Quick check:** in 2018 Notre Dame gained 550 yards against Stanford. What does this line predict, and what is the residual if Notre Dame scored 38?

## Linear regression visualization



# Representing curves

How could you use a **linear regression** to model a sine wave?




The model only has to be linear in its _____, not in its inputs.

# A possible issue

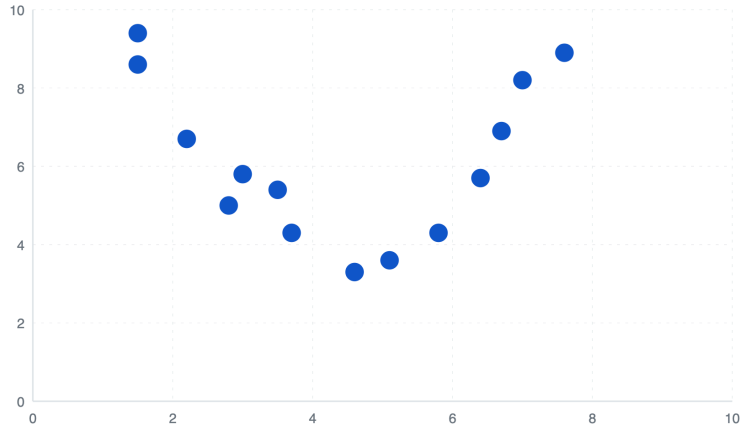
The **weight calculation relies on computing the inverse**, which can be **extremely computationally expensive** and **sometimes just literally impossible**.

# Maximum likelihood estimation

**Maximum likelihood estimation (MLE)** is a method of estimating the parameters of an _____, **given some observed data**.

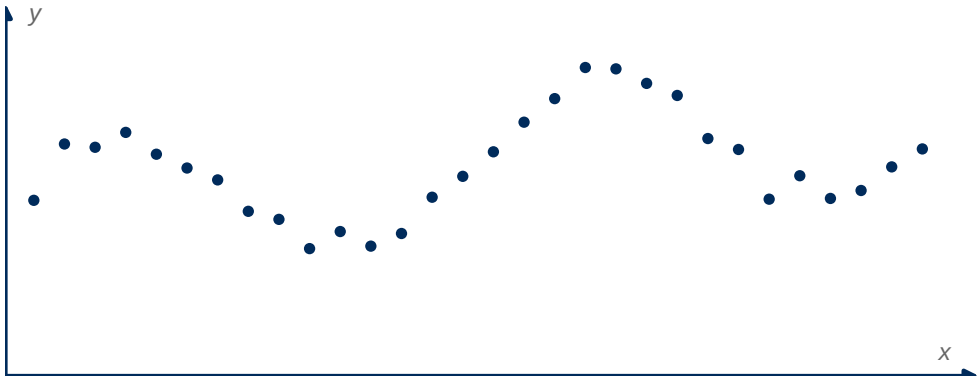
## Overfitting

Unfortunately, **Linear Regressions** (a form of MLE), are **prone to overfitting**.

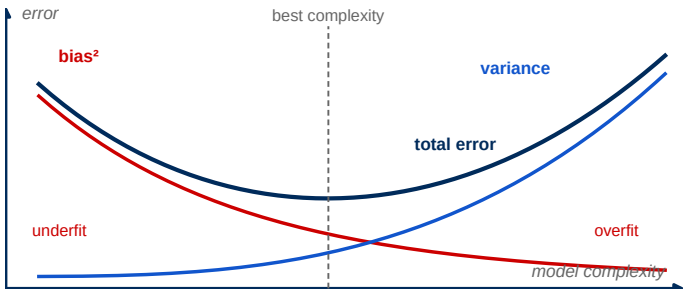


# Underfitting

**Underfitting** is when your model is "too weak" to capture all of the underlying patterns in the data.




## Bias and variance




The **bias–variance tradeoff** describes the **relationship between** a **model's complexity**, the **accuracy** of its predictions, and how well it can make **predictions on unseen data** that were not used to train the model.

# Practice: an old exam question

| $x_1$ | $x_2$ | $y$ |
|---|---|---|
| 1 | 1 | 9 |
| 2 | 2 | 25 |
| 3 | 3 | 50 |
| 4 | 4 | 84 |
| 5 | 5 | 127 |

Given the dataset above, answer the following questions **(Old exam question)**:

**A.** Would a **purely linear model** exhibit **high bias** here or **high variance**? Would it be **overfit** or **underfit**? Explain your reasoning.

**B.** Design a regression equation that would be able to approximate the dataset.

**C.** What role does the **SSE** play in **linear regression**? Why does it matter that our resulting parameter space is convex?

**D.** Look at the $x_1$ and $x_2$ columns. What goes wrong if we fit $\hat{y} = w_0 + w_1 x_1 + w_2 x_2$ with **OLS**?

# Matching: linear regression

Write the letter of the matching definition next to each term.

| | |
|---|---|
| ___ **Ordinary Least Squares** | **A.** The model memorizes noise in the training data instead of the general pattern |
| ___ **Maximum Likelihood Estimation** | **B.** The relationship between model complexity, training accuracy, and accuracy on unseen data |
| ___ **Underfitting** | **C.** Solving for the weights by setting the gradient of the loss equal to 0 |
| ___ **Overfitting** | **D.** Estimating the parameters that make the observed data most likely |
| ___ **Bias–variance tradeoff** | **E.** The model is too weak to capture the underlying patterns in the data |